

The SEMICONDUCTOR INDUSTRY Has ROOM For EVERYONE Whether Big Or Small

There has been more than enough talk around setting up a semiconductor manufacturing ecosystem in India but almost zero discussion on whether the small and medium sized companies can fit in or not. If these can fit in, what are the opportunities available, and what's the best way to grab them!

MUKUL YUDHVEER SINGH

“Semiconductors are a very big opportunity today. They are the key to everything electronic now. The vertical is becoming more and more important because electronics are becoming more and more intelligent. ELCINA is of view that India needs more ATMPs in the country to be able to take Fabs forward. I feel that ATMPs and OSATs will help us attract Fabs in India,” says Rajoo Goel, secretary general, ELCINA.

The government of India, has recently announced a programme for promoting semiconductor manufacturing in the country. This programme, worth over ₹760 billion, aims to not only provide incentives to semiconductor fabs but also to organisations present in the semiconductor design. The move seems brilliantly timed as the world is currently haunted by the shortage of semiconductors. Moreover, as the semiconductors are the brains behind modern electronics, they are of strategic value to all the countries.

“It’s a fact that the electronics industry was crushed in the early part of the millen-

nium. The situation now, thanks to the government’s bold PLI and SPECS programme, is changing for good. India’s electronics industry has shown tremendous growth. We have seen the end product EMS come up. It has the capability of helping establish a better ecosystem of manufacturing components in the country. This is a catalytic reaction that will drive the growth of components and as well as end products in India,” shares Dr Ashwini K. Aggarwal, PhD, FIETE, Director, Applied Materials India.

Semiconductors, as a matter of fact, have been on the radar of the government of India for a long time now. The government of India has admitted that in the current geopolitical scenario, trusted sources of semiconductors and displays hold strategic importance and are key to the security of critical information infrastructure.

“Semiconductor manufacturing has been a mirage for India for the past several decades. India’s semiconductor aspirations have remained unsatiated despite the magnitude of opportunities possible. When we stand as a nation and say we will, then we will,” says Ashwini.

Opportunity in numbers

The semiconductor industry, as per Applied Materials, will be worth over \$1 trillion in 2030. Industry 4.0 is being termed as one of the major factors driving this growth. As a matter of fact, machines had already surpassed the amount of data being generated by humans in the year 2018 itself. Seagate UK estimates that by 2025 there will be 175 zettabytes of data in the global datasphere.

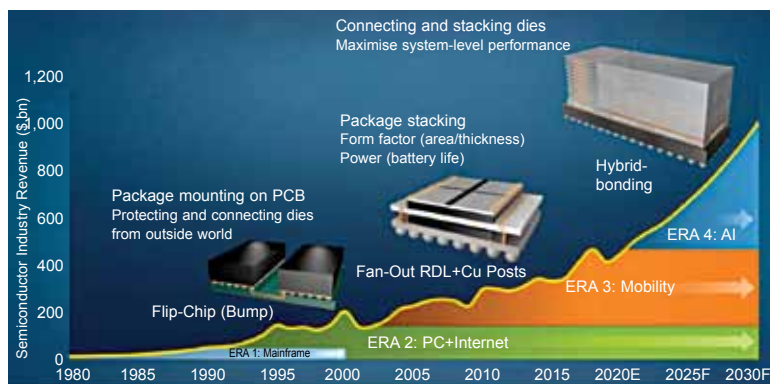


Fig. 1: Semiconductor packaging